

Vegetarian diets are associated with healthy mood states: a cross-sectional study in Seventh Day Adventist adults

Background The physical health status of vegetarians has been extensively reported, but there is limited research regarding the mental health status of vegetarians, particularly with regard to mood. Vegetarian diets exclude fish, the major dietary source of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), critical regulators of brain cell structure and function. Omnivorous diets low in EPA and DHA are linked to impaired mood states in observational and experimental studies. **Methods** We examined associations between mood state and polyunsaturated fatty acid intake as a result of adherence to a vegetarian or omnivorous diet in a cross-sectional study of 138 healthy Seventh Day Adventist men and women residing in the Southwest. Participants completed a quantitative food frequency questionnaire, Depression Anxiety Stress Scale (DASS), and Profile of Mood States (POMS) questionnaires. **Results** Vegetarians (VEG:n = 60) reported significantly less negative emotion than omnivores (OMN:n = 78) as measured by both mean total DASS and POMS scores (8.32 ± 0.88 vs 17.51 ± 1.88 , $p = .000$ and 0.10 ± 1.99 vs 15.33 ± 3.10 , $p = .007$, respectively). VEG reported significantly lower mean intakes of EPA ($p < .001$), DHA ($p < .001$), as well as the omega-6 fatty acid, arachidonic acid (AA; $p < .001$), and reported higher mean intakes of shorter-chain α -linolenic acid ($p < .001$) and linoleic acid ($p < .001$) than OMN. Mean total DASS and POMS scores were positively related to mean intakes of EPA ($p < 0.05$), DHA ($p < 0.05$), and AA ($p < 0.05$), and inversely related to intakes of ALA ($p < 0.05$), and LA ($p < 0.05$), indicating that participants with low intakes of EPA, DHA, and AA and high intakes of ALA and LA had better mood. **Conclusions** The vegetarian diet profile does not appear to adversely affect mood despite low intake of long-chain omega-3 fatty acids.